

# A. Feder Cooper, Ph.D.

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I research a variety of topics in reliable, scalable machine learning. My contributions span privacy, security, evaluation of generative-AI systems, MLSys, and measurement validity. I also do work in tech policy and law, and spend a lot of time finding ways to effectively communicate the capabilities and limits of AI/ML to interdisciplinary audiences and the public. My research has received spotlights, orals, and best-paper accolades at top AI/ML and computing venues, including *NeurIPS*, *ICML*, *AAAI*, and *AIES*. Collaborations on copyright and generative AI have been lauded as “landmark” work among scholars and the international press.

## Positions and Appointments

Assistant Professor, Department of Computer Science, Yale University (starting July 2026)

Co-lead, The GenLaw Center, July 2023–present

Founding Research Scientist, AI Verification and Evaluation Research Institute (AVERI)

Postdoctoral Affiliate, Stanford HAI, RegLab, and CRFM

Faculty Associate at the Berkman Klein Center for Internet & Society at Harvard University

## Education

### Cornell University

Ph.D., Computer Science

2018 – 2024

[“Between Randomness and Arbitrariness: Some Lessons for Reliable Machine Learning at Scale”](#)

M.S., Computer Science

2021

### Columbia University

B.A., Computer Science and Archaeology, Phi Beta Kappa, *summa cum laude*

2014

## Selected Papers

\*Denotes co-first author or equal contribution; †lead author (not specified if same as first author); [Google Scholar](#)

**A. Feder Cooper**, Mark A. Lemley, Christopher De Sa, Lea Duesterwald, Allison Casasola, Jamie Hayes, Katherine Lee, Daniel E. Ho, and Percy Liang. “Estimating near-verbatim extraction risk in language models with decoding-constrained beam search.” *ICML ’26 Workshop on the Impact of Memorization on Trustworthy Foundation Models*. [\[link\]](#) [Workshop](#) [Oral](#)

Miles Brundage, Noemi Dreksler, Aidan Homewood, Sean McGregor, Patricia Paskov, Conrad Stosz, Girish Sastry, **A. Feder Cooper**, George Balston, Steven Adler, Stephen Casper, Markus Anderljung, Grace Werner, Soren Mindermann, Vasilios Mavroudis, Ben Bucknall, Charlotte Stix, Jonas Freund, Lorenzo Pacchiardi, Jose Hernandez-Orallo, Matteo Pistillo, Michael Chen, Chris Painter, Dean W. Ball, Cullen O’Keefe, Gabriel Weil, Ben Harack, Graeme Finley, Ryan Hassan, Scott Emmons, Charles Foster, Anka Reuel, Bri Treece, Yoshua Bengio, Daniel Reti, Rishi Bommasani, Cristian Trout, Ali Shahin Shamsabadi, Rajiv Dattani, Adrian Weller, Robert Trager, Jaime Sevilla, Lauren Wagner, Lisa Soder, Ketan Ramakrishnan, Henry Papadatos, Malcolm Murray, and Ryan Tovicimak. “Frontier AI Auditing: Toward Rigorous Third-Party Assessment of Safety and Security Practices at Leading AI Companies.” 2026. [\[link\]](#) [arXiv](#)

Ahmed Ahmed\*, **A. Feder Cooper**\*, Sanmi Koyejo, and Percy Liang. “Extracting books from production language models.” 2026. [\[link\]](#) [arXiv](#)

**A. Feder Cooper**, Mark A. Lemley, Allison Casasola, Ahmed Ahmed, Aaron Gokaslan, Amy B. Cyphert, Christopher De Sa, Daniel E. Ho, Percy Liang. “Extracting memorized pieces of (copyrighted) books from open-weight language models.” *ICML ’25 Workshop on Reliable and Responsible Foundation Models*. [\[link\]](#) [Workshop](#) [Oral](#)  
(≈3% of submitted papers)

Jamie Hayes\*, Ilia Shumailov, Christopher A. Choquette-Choo, Matthew Jagielski, George Kaissis, Milad Nasr, Meenatchi Sundaram Mutu Selva Annamalai, Niloofar Mireshghallah, Igor Shilov, Matthieu Meeus, Yves-Alexandre de Montjoye, Katherine Lee, Franziska Boenisch, Adam Dziedzic, and **A. Feder Cooper\***. “Exploring the limits of strong membership inference attacks on large language models.” *NeurIPS* ’25. [[link](#)] **Proceedings** (24.52% acceptance)

Nikhil Kandpal\*, Brian Lester\*, Colin Raffel\*, Sebastian Majstorovic, Stella Biderman, Baber Abbasi, Luca Soldaini, Enrico Shippole, **A. Feder Cooper**, Aviya Skowron, John Kirchenbauer, Shayne Longpre, Lintang Sutawika, Alon Albalak, Zhenlin Xu, Guilherme Penedo, Loubna Ben Allal, Elie Bakouch, John David Pressman, Honglu Fan, Dashiell Stander, Guangyu Song, Aaron Gokaslan, Tom Goldstein, Brian R. Bartoldson, Bhavya Kailkhura, and Tyler Murray. “The Common Pile v0.1: An 8TB Dataset of Public Domain and Openly Licensed Text.” *NeurIPS* ’25, *Datasets and Benchmarks Track*, [[link](#)] **Proceedings** (24.91% acceptance)

**A. Feder Cooper\***<sup>†</sup>, Christopher A. Choquette-Choo\*, Miranda Bogen\*, Kevin Klyman\*, Matthew Jagielski\*, Katja Filippova\*, Ken Ziyu Liu\*, Alexandra Chouldechova, Jamie Hayes, Yangsibo Huang, Eleni Triantafillou, Peter Kairouz, Nicole Mitchell, Niloofar Mireshghallah, Abigail Z. Jacobs, James Grimmelmman, Vitaly Shmatikov, Christopher De Sa, Ilia Shumailov, Andreas Terzis, Solon Barocas, Jennifer Wortman Vaughan, danah boyd, Yejin Choi, Sanmi Koyejo, Fernando Delgado, Percy Liang, Daniel E. Ho, Pamela Samuelson, Miles Brundage, David Bau, Seth Neel, Hanna Wallach, Amy B. Cyphert, Mark A. Lemley, Nicolas Papernot, and Katherine Lee\*<sup>†</sup>. “Machine Unlearning Doesn’t Do What You Think: Lessons for Generative AI Policy and Research.” *NeurIPS* ’25, *Position Track* [[link](#)] **Proceedings** **Oral** (≈0.4% of submitted papers).

Alexandra Chouldechova\*, **A. Feder Cooper\***, Solon Barocas, Abhinav Palia, Dan Vann, and Hanna Wallach. “Comparison requires valid measurement: Rethinking attack success rate comparisons in AI red teaming.” *NeurIPS* ’25, *Position Track* [[link](#)] **Proceedings** (5.7% acceptance)

Hanna Wallach, Meera Desai, **A. Feder Cooper**, Angelina Wang, Solon Barocas, Alexandra Chouldechova, Chad Atalla, Su Lin Blodgett, Emily Corvi, P. Alexander Dow, Jean Garcia-Gathright, Alexandra Olteanu, Nicholas J. Pangakis, Stefanie Reed, Emily Shang, Dan Vann, Jenn Wortman Vaughan, Matthew Vogel, Hannah Washington, and Abigail Z. Jacobs. “Evaluating Generative AI Systems is a Social Science Measurement Challenge.” *ICML* ’25, *Position Track* [[link](#)] **Proceedings** (19.7% acceptance)

Jamie Hayes\*, Marika Swanberg, Harsh Chaudhari, Itay Yona, Ilia Shumailov, Milad Nasr, Christopher A. Choquette-Choo, Katherine Lee, and **A. Feder Cooper\***. “Measuring memorization in language models via probabilistic extraction.” *NAACL* ’25 [[link](#)] **Proceedings** (≈22% acceptance)

**A. Feder Cooper\*** and James Grimmelmman\*. “The Files are in the Computer: On Copyright, Memorization, and Generative AI.” *Chicago-Kent Law Review*, Vol. 100, 2025. [[link](#)] **Journal** (written 2024)

Katherine Lee\*, **A. Feder Cooper\***, and James Grimmelmman\*. “Talkin’ ’Bout AI Generation: Copyright and the Generative-AI Supply Chain.” *Journal of the Copyright Society*, Volume 72, 2025. [[link](#)] **Journal** (written 2023)

Milad Nasr\*, Javier Rando\*, Nicholas Carlini\*, Jonathan Hayase, Matthew Jagielski, **A. Feder Cooper**, Daphne Ippolito, Christopher A. Choquette-Choo, Florian Tramèr, and Katherine Lee<sup>†</sup>. “Scalable Extraction of Training Data from Aligned, Production Language Models.” *ICLR* ’25 [[link](#)] **Proceedings** (32% acceptance)

Nicholas Carlini, Daniel Paleka, Krishnamurthy Dj Dvijotham, Thomas Steinke, Jonathan Hayase, **A. Feder Cooper**, Katherine Lee, Matthew Jagielski, Milad Nasr, Arthur Conmy, Eric Wallace, David Rolnick, and Florian Tramèr. “Stealing Part of a Production Language Model.” *ICML* ’24. [[link](#)] **Proceedings** **Best Paper** (≈0.1% of submitted papers)

**A. Feder Cooper**, Katherine Lee, Madiha Zahrah Choksi, Solon Barocas, Christopher De Sa, James Grimmelmman, Jon Kleinberg, Siddhartha Sen, and Baobao Zhang. “Arbitrariness and Social Prediction: The Confounding Role of Variance in Fair Classification.” *AAAI* ’24 [[link](#)] **Proceedings** **Best Student Paper Honorable Mention** (<1% of submitted papers; written 2023)

**A. Feder Cooper\***<sup>†</sup>, Wentao Guo\*, Khiem Pham\*, Tiancheng Yuan, Charlie F. Ruan, Yucheng Lu, and Christopher De Sa<sup>†</sup>. “Coordinating Distributed Example Orders for Provably Accelerated Training.” *NeurIPS* ’23 [[link](#)] **Proceedings** (26% acceptance)

Milad Nasr\*, Nicholas Carlini\*, Jonathan Hayase, Matthew Jagielski, **A. Feder Cooper**, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, and Katherine Lee†. “Scalable Extraction of Training Data from (Production) Language Models.” 2023. [\[link\]](#) [arXiv](#)

**A. Feder Cooper**\*†, Katherine Lee\*, James Grimmelmänn\*, Daphne Ippolito\*, Christopher Callison-Burch, Christopher A. Choquette-Choo, Niloofar Miresghallah, Miles Brundage, David Mimno, Madiha Zahrah Choksi, Jack M. Balkin, Nicholas Carlini, Christopher De Sa, Jonathan Frankle, Deep Ganguli, Bryant Gipson, Andres Guadamuz, Swee Leng Harris, Abigail Z. Jacobs, Elizabeth Joh, Gautam Kamath, Mark Lemley, Cass Matthews, Christine McLeavey, Corynne McSherry, Milad Nasr, Paul Ohm, Adam Roberts, Tom Rubin, Pamela Samuelson, Ludwig Schubert, Kristen Vaccaro, Luis Villa, Felix Wu, and Elana Zeide. “Report of the 1st Workshop on Generative AI and Law.” 2023. [\[link\]](#) [arXiv](#)

**A. Feder Cooper**, Jonathan Frankle, and Christopher De Sa. “Non-Determinism and the Lawlessness of Machine Learning Code.” *CSLAW ’22* [\[link\]](#) [Proceedings](#)

**A. Feder Cooper** and Karen Levy. “Fast or Accurate? Governing Conflicting Goals in Highly Autonomous Vehicles.” *Colorado Technology Law Journal*, Volume 20, 2022. [\[link\]](#) [Journal](#)

**A. Feder Cooper**\*†, Emanuel Moss\*, Benjamin Laufer, and Helen Nissenbaum†. “Accountability in an Algorithmic Society: Relationality, Responsibility, and Robustness in Machine Learning.” *FACCT ’22* [\[link\]](#) [Proceedings](#) (25% acceptance)

**A. Feder Cooper**, Yucheng Lu, Jessica Zosa Forde, and Christopher De Sa. “Hyperparameter Optimization Is Deceiving Us, and How to Stop It.” *NeurIPS ’21* [\[link\]](#) [Proceedings](#) (<26% acceptance)

Ruqi Zhang\*, **A. Feder Cooper**\*, and Christopher De Sa†. “Asymptotically Optimal Exact Minibatch Metropolis-Hastings.” *NeurIPS ’20* [\[link\]](#) [Proceedings](#) [Spotlight](#) (<3% of submitted papers)

## Selected Honors, Grants, and Awards

Stanford HAI Seed Grant (\$75,000), “Assessing Copyright Risks of Training Data ‘Memorization’ and ‘Extraction’ for Open-Weight Language Models” (2026 – 2027)

*NeurIPS* Oral, [Machine Unlearning Doesn’t Do What You Think: Lessons for Generative AI Policy and Research](#) (≈0.4% of submitted papers) (2025)

*ICML* Best Paper Award, [Stealing Part of a Production Model](#) (10 of 9473 (≈0.1%) submitted papers) (2024)

*AAAI* Best Student Paper Honorable Mention, [Arbitrariness and Social Prediction: The Confounding Role of Variance in Fair Classification](#) (<1% of submitted papers) (2024)

Meta Ph.D. Fellowship Finalist (2023) (short-listed out of 3200 candidates)

Rising Star in EECS, MIT (2021)

Fellowship Finalist, OpenPhil AI Ph.D. Fellowship (2021)

## Conference, Workshop, and Tutorial Organization

**5th ACM Symposium on Computer Science and Law (CS&Law ’26)**. Executive committee member, publicity chair. Held at UC Berkeley (Berkeley, CA). [\[link\]](#)

**Data Privacy, Memorization, & Legal Implications in Generative AI: A Practical Guide**. *NeurIPS ’25* Tutorial, December 2, 2025 (San Diego, CA).

**4th ACM Symposium on Computer Science and Law (CS&Law ’25)**. Executive committee member, publicity co-chair. Held at TU Munich (Munich, Germany). [\[link\]](#)

**2nd Workshop on Generative AI and Law (GenLaw ’24)**. Held at *ICML 2024* in Vienna, Austria. Lead co-organizer with Katherine Lee. [\[link\]](#)

**Evaluating Generative AI Systems: the Good, the Bad, and the Hype (GenLaw DC '24).** Lead co-organizer. Sponsored by the K&L Gates initiative in Ethics and Computational Technologies at Carnegie Mellon University (CMU), and organized by the GenLaw Center (A. Feder Cooper, Katherine Lee, James Grimmelmann), Carnegie Mellon's K&L Gates Initiative (Hoda Heidari), the Georgetown Law Center (Paul Ohm), and the Center for Democracy and Technology (Alexandra Reeve Givens, Miranda Bogen). [[link](#)]

**1st Workshop on Generative AI and Law (GenLaw '23).** Held at *ICML 2023* in Honolulu, Hawai'i, 26% acceptance rate. Lead co-organizer with Katherine Lee; co-organized with Dr. Niloofar Mireshghallah, Madiha Zahrah Choksi, Prof. James Grimmelmann, Prof. David Mimno, and Dr. Deep Ganguli [[link](#)]

## Selected Recent Talks

Invited participant, Symposium on AI and Law, Harvard Law School, October 24, 2025 (Cambridge, MA).

"What Copyright Can Learn from Memorization Measurements of Language Models."

Invited talk, American National Standards Institute (ANSI), May 11, 2026 (Virtual).

Invited speaker, Workshop & Lecture Series on the Law & Economics of Innovation, ETH Zürich, October 28–29, 2025 (Zürich, Switzerland).

Invited speaker, Information Technology and Innovation Foundation, August 2025 (Virtual).

Invited speaker, [Workshop on the Impact of Memorization on Trustworthy Foundation Models](#), *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

Invited panelist, [Workshop on Machine Unlearning for Generative AI](#), *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

AI Governance Tutorial (with Katherine Lee, 28th Annual BTLJ-BCLT Spring Symposium: AI Governance at the Crossroads, February 27, 2025 (Berkeley, CA).

"Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice."

Future of Privacy Forum, Technologist Roundtable for Policymakers, July 2025 (Virtual).

Guest Lecture for CPSC 310: Technology, Power, and Security: Political Challenges of the Computer Age, Yale University, April 8, 2025 (New Haven, CT).

Invited talk, NYU Law Innovation Policy Colloquium, April 3, 2025 (New York, NY).

Future of Privacy Forum, January 13, 2025 (Virtual – Washington, DC).

## Selected Service

Area Chair for *COLM* (2025, 2026), *ICML* (2026), *NeurIPS* (2026)

Publicity co-chair (with Aloni Cohen) for *CSLAW* (2025); publicity chair for *CSLAW* (2026)

Reviewer / Program Committee for *NeurIPS* (2020-2024), *ICML* (2021-2023, 2025), *ICLR* (2022-2023, 2025-2026), *TMLR* (2022-present), *AAAI* (2025), *CSLAW* (2025–2026), *SATML* (2025), *ACM CACM* (2025-2026), *COLM* (2024), *FAccT* (2021-2023), *JMLR* (2022), *AIES* (2021), *EAAMO* (2021)